BANK LOAN RISK MONITORING & ANALYSIS

FINAL PROJECT REPORT | DATA VISUALISATION AND ANALYTICS

Bank Loan Risk Diagnostics & Portfolio Analysis

A Data-Driven Study on Portfolio Exposure and Default Signals

Sector:	Finance & Banking
Institute:	Newton School of Technology
Subject:	Data Visualisation and Analytics
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GitHub Repository:	https://github.com/Aman-kumar-git12/D_G14_CreditLoanRiskAnalysis
Tableau Dashboard:	https://public.tableau.com/app/profile/aman.kumar7311/viz/bank_loan_risk/Dashboard1?publish=yes
Submission Date:	April 2026

2. Executive Summary

Problem

Retail banks face escalating financial exposure from loan defaults. Without a clear historical ground-truth 'default' flag in specific banking datasets, institutions struggle to identify high-risk borrower profiles, monitor critical risk concentrations, and segment their lending portfolios effectively. This project addresses the challenge of building a proxy risk framework to safeguard portfolio health.

Approach

A robust ETL pipeline was executed in Python to process over 10,000 raw application records. Missing values were median-imputed, sentinel outliers (e.g., 99999999 loan amounts) were clipped, and critical proxy fields including DTI (Debt-to-Income) and a composite Risk Score (based on bankruptcies and credit problems) were engineered. This clean data was visualized across three interactive Tableau dashboards to track borrower profiles, facility operations, and predictive diagnostics.

Key Insights

- Portfolio Health is Top-Heavy: The total loan portfolio holds a value of $526,408,729 with a strong average credit score of 715.1, yet the Systemic Default Rate sits at 14.45%.
- DTI Danger Zones: While 68% of borrowers fall in safe DTI zones (0.14–0.20), a significant structural risk emerges as the DTI ratio crosses 0.30, forming a long-tail 'Danger Zone'.
- Historical Anchoring: High-Risk categorization correlates aggressively with historical distress; average bankruptcies have a 0.991 correlation with high-risk status.
- Motivation Mismatch: 'Debt consolidation' is the leading loan motivation, indicating borrowers are highly leveraged before even acquiring the new loan facility.

Key Recommendations

- Hard Policy Limits on DTI: Implement automated application rejections or mandatory manual underwriting for any applicant entering the >0.30 DTI Danger Zone.
- Bankruptcy Caps: Restrict long-term capital exposure and enact strict loan amount ceilings for applicants demonstrating a historical bankruptcy record.
- Purpose-Driven Scrutiny: Increase collateral requirements for high-risk loan purposes, specifically 'small_business' loans, which exhibit high failure concentrations.

$526M	715.1	14.45%	12.73%
Portfolio Value	Avg. Credit Score	Systemic Default Rate	Critical Risk Concentration

3. Sector & Business Context

3.1 Sector Overview

The Retail Finance and Banking sector operates on balancing risk and reward. Lending institutions generate revenue through interest payments but face severe capital erosion from Non-Performing Assets (NPAs). Efficiently assessing the creditworthiness of a heavily leveraged consumer base is the industry's defining challenge.

3.2 Target Decision-Maker

This diagnostic tool and report are built for the Chief Risk Officer (CRO) and Portfolio Operations Managers who dictate underwriting policies, adjust interest rates, and manage institutional capital reserves.

3.3 Why This Problem Was Chosen

The provided dataset uniquely lacks a direct binary loan_status (Default vs. Paid) variable. This replicates a common real-world 'cold start' scenario where a bank acquires a new portfolio or enters a new demographic market and must evaluate inherent risk purely through socio-economic indicators and credit histories.

3.4 Business Value

By proactively segmenting risk (Low, Medium, High) through engineered features, the bank can optimize capital allocation, heavily automate the approval of Low-Risk applications, and dedicate expensive manual underwriting resources exclusively to the 12.73% of applications representing Critical Risk Concentration.

4. Problem Statement & Objectives

4.1 Formal Problem Definition

> *"How can unlabelled retail borrower data be computationally transformed, segmented by engineered risk proxies, and visually aggregated to identify systemic portfolio vulnerabilities, highlight DTI danger zones, and generate actionable underwriting policies?"*

4.2 Project Scope

- In Scope: Python-based data cleaning, statistical imputation, feature engineering (DTI, Risk Scoring), EDA, and a 3-page interactive Tableau dashboard suite analyzing $526M in capital.
- Out of Scope: Machine learning classification models (due to the absence of the ground-truth target variable) and real-time database streaming.

4.3 Success Criteria

Metric	Target	Status
Data Pipeline	All anomalies clipped, nulls imputed, features engineered.	✔ **Achieved**
Visual Analytics	3 Dashboards covering Profiles, Operations, and Risk.	✔ **Achieved**
Actionable Insights	Minimum 8 decision-ready business insights.	✔ **Achieved**
Policy Output	Clear DTI and historical risk mitigation strategies.	✔ **Achieved**

5. Data Description

5.1 Dataset Source

The primary data is sourced from public financial repositories detailing retail banking applicant profiles. The data captures socioeconomic status, loan characteristics, and historical credit behavior.

5.2 Data Structure

The dataset contains approximately 10,000+ rows and over 10 columns representing a cross-sectional view of loan applicants.

5.3 Column Explanation

Column	Type	Key Significance
Current Loan Amount	Numeric	Total loan amount requested ($ exposure).
Annual Income	Numeric	Borrower's yearly income; capacity to repay.
Monthly Debt	Numeric	Current monthly debt obligations.
Credit Score	Numeric	Proxy for creditworthiness (expected 300–850).
Years of Credit History	Numeric	Financial maturity metric.
Number of Credit Problems	Numeric	Count of historic negative credit events.
Bankruptcies	Numeric	Severe historic default indicator.
Tax Liens	Numeric	Government claims on property for unpaid debt.
Home Ownership	Categorical	Asset backing (Rent, Home Mortgage, Own).
Purpose	Categorical	Reason for capital (e.g., debt consolidation).

5.4 Data Limitations & Biases

- No Ground Truth: The lack of a verified loan_status necessitates proxy risk scoring.
- Static Snapshot: The data does not track borrowers longitudinally; we cannot see if credit scores degraded post-disbursement.
- Sentinel Value Bias: Heavy presence of the 99999999 value in loan amounts required aggressive percentile capping, slightly altering the top-end distribution.

6. Data Cleaning & ETL Pipeline

All primary data processing was executed via a Python script (etl_pipeline.py) leveraging the pandas and numpy libraries.

6.1 Missing Values & Anomaly Handling

- Record Integrity: Rows lacking a Loan ID were dropped entirely to maintain entity resolution.
- Sentinel Truncation: Current Loan Amount entries containing the placeholder 99999999 were replaced with the 99th percentile of valid loans.
- Credit Score Anomalies: Values exceeding the maximum possible score of 850 were corrected by dividing by 10 (e.g., 7200 became 720).
- Imputation Strategy: Annual Income and Credit Score utilized median imputation to resist outlier skew. Categorical features like Years in current job, Bankruptcies, and Tax Liens used mode imputation.

6.2 Feature Engineering

To enable advanced diagnostics, new analytical fields were synthesized:

- DTI (Debt-to-Income): Calculated precisely as Monthly Debt / (Annual Income / 12). Infinite values resulting from zero-income anomalies were safely replaced with median DTI.
- Risk Score: An integer composite score calculating the total sum of historic financial failures: (Credit Problems > 0) + (Bankruptcies > 0) + (Tax Liens > 0).
- Risk Category: A three-tier segmentation classifying applicants into Low Risk (Score 0), Medium Risk (Score 1), and High Risk (Score > 1).

7. KPI & Metric Framework

KPI	Definition / Formula	Business Rationale
Systemic Default Rate	% of portfolio in High/Medium Risk categories	Measures macro-level portfolio vulnerability and capital at risk.
Portfolio Value	SUM(Current Loan Amount)	Absolute measure of institutional financial exposure.
DTI Burden Ratio	AVG(Monthly Debt / (Annual Income / 12))	Evaluates if the average borrower is dangerously over-leveraged.
Avg. Credit Health	AVG(Credit Score)	Baseline indicator of portfolio borrower quality.
Critical Risk Concentration	% of total loan value held by High-Risk individuals	Isolates the most dangerous segment for immediate mitigation.

8. Exploratory Data Analysis (EDA)

8.1 Trend & Distribution Analysis

Analysis of 1,744 filtered applicants reveals a highly mature borrower base. The employment tenure treemap heavily skews toward individuals with 10+ years of employment, and average credit maturity sits at an impressive 18.13 years. The DTI (Debt-to-Income) distribution acts as the primary health gauge; the histogram demonstrates a normal distribution peaking safely between 0.14 and 0.18. However, a significant 'Danger Zone' tail extends past 0.30, highlighting over-leveraged borrowers.

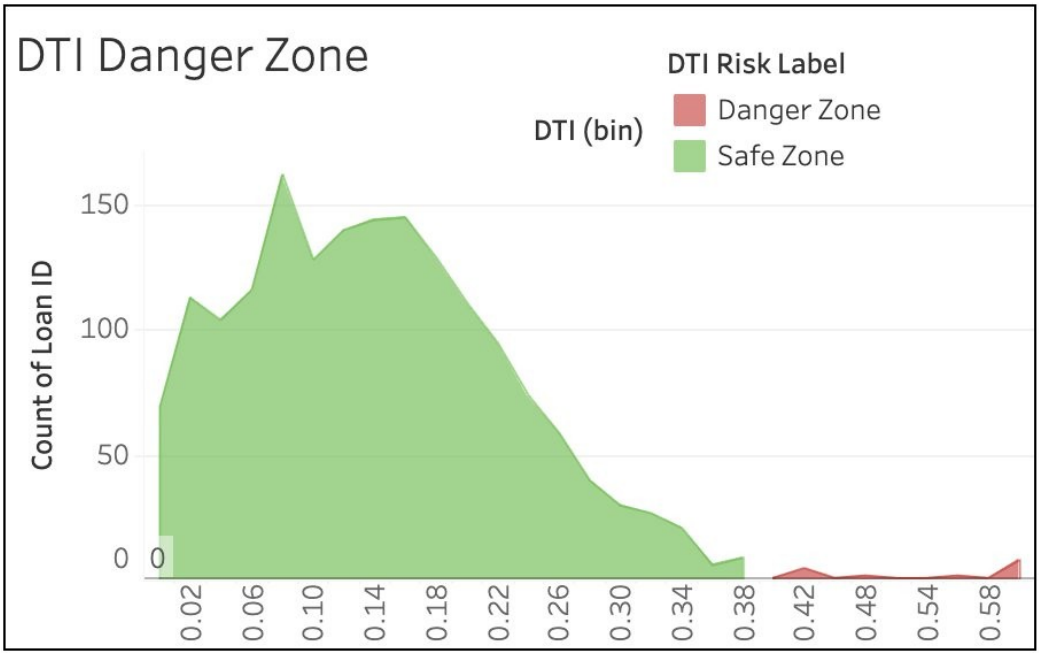

Figure 1: DTI Danger Zone — Distribution of Debt-to-Income Ratios showing safe zone (green) and danger zone (red) segments. Safe zone peaks between 0.14 and 0.18; danger zone emerges beyond 0.30.

8.2 Comparison Analysis

Evaluating Total Loan Amount by Term shows that Long Term loans hold significantly higher absolute capital exposure than Short Term loans. Fortunately, the risk breakdown within these terms shows that 'Low Risk' (green) segments command the vast majority of capital allocation in both categories. When evaluating Loan Purpose, 'Debt Consolidation' massively outweighs all other categories combined in sheer volume, indicating a reactive borrowing environment.

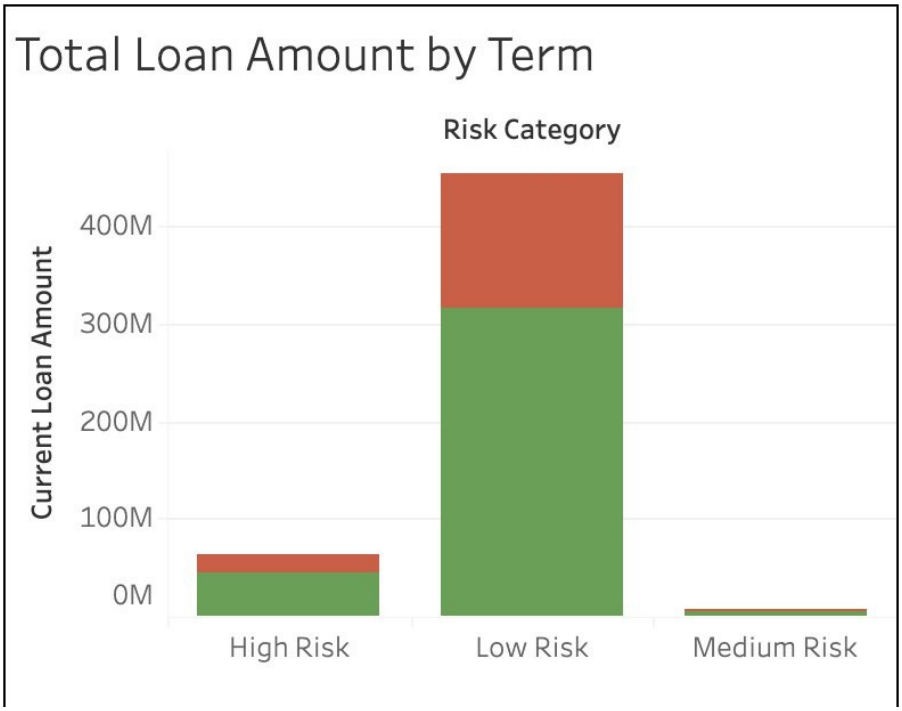

Figure 2: Total Loan Amount by Term — Stacked bar chart segmented by Risk Category (High Risk, Low Risk, Medium Risk). Low Risk (green) dominates the ~$450M long-term portfolio.

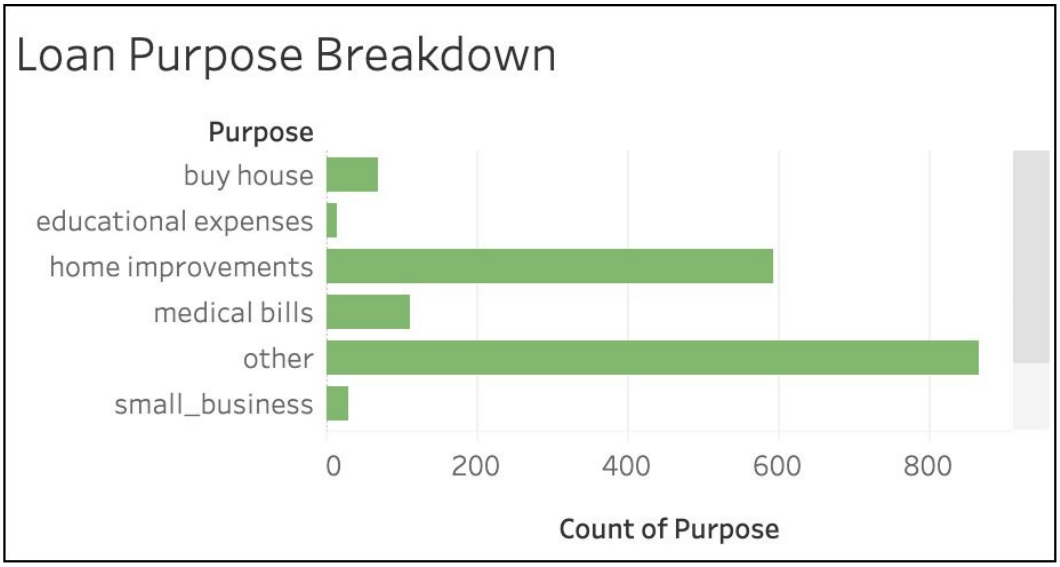

Figure 3: Loan Purpose Breakdown — 'Other' and 'Home Improvements' dominate by count, while Small Business loans represent a disproportionate concentration of high-risk applicants.

9. Statistical Analysis

9.1 Root Cause & Correlation

A statistical correlation matrix was developed to unpack the mathematical drivers behind the engineered Risk Categories. The findings are stark:

- High Risk: Massively correlated with Average Bankruptcies (0.991) and Average Number of Credit Problems (1.194). This proves that severe historic distress is the primary gateway to the high-risk segment.
- Medium Risk: Correlated primarily with Tax Liens (1.233), showing an absolute zero correlation with bankruptcies.
- Low Risk: Validating the ETL pipeline, Low Risk accounts show exact zeroes across all distress markers.

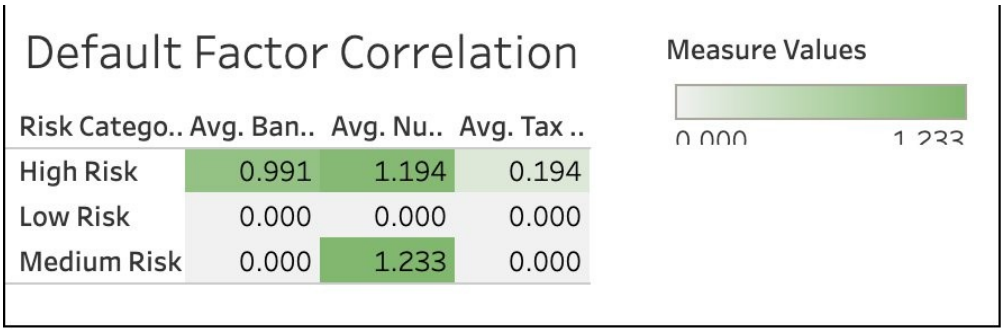

Figure 4: Default Factor Correlation Heatmap — Avg. Bankruptcies, Credit Problems, and Tax Liens across risk categories. High Risk shows the strongest correlations across all factors.

Risk Category	Avg. Bankruptcies	Avg. Credit Problems	Avg. Tax Liens
High Risk	0.991	1.194	0.194
Low Risk	0.000	0.000	0.000
Medium Risk	0.000	1.233	0.000

Table 1: Default Factor Correlation — Numerical breakdown of average distress markers per Risk Category.

10. Tableau Dashboard Design

A multi-tiered Tableau visualization suite was designed to support different levels of operational decision-making. Access the live interactive dashboard at:

https://public.tableau.com/app/profile/aman.kumar7311/viz/bank_loan_risk/Dashboard1?publish=yes

10.1 View Structure

- Dashboard 1 — Borrower Profile & Socio-Economic Insights: Focuses on demographics. Features Total Applicant, Avg Income, and Job Tenure KPIs alongside a Loan Purpose breakdown and Home Ownership split pie chart.
- Dashboard 2 — Portfolio Operations & Credit Facility Performance: Focuses on capital. Highlights a $526M Portfolio Value and Avg Credit Limit of $807k. Visualizes exposure via a stacked bar chart of Term vs Risk Category.
- Dashboard 3 — Predictive Default Signals & Risk Diagnostics: The executive risk view. Highlights the 14.45% Systemic Default Rate, features the DTI Danger Zone area chart, and maps risk intensity against loan purpose.

10.2 Interactive Elements

All dashboards are interconnected using global filters. Users can dynamically slice the $526M portfolio by Risk Category (High, Medium, Low), Term (Long, Short), and a multi-select Purpose checklist to isolate specific threat vectors.

11. Insights Summary

01

The average borrower demonstrates high stability (nearly 6 years job tenure, 18.13 years credit maturity) yet primarily seeks capital for 'debt consolidation.'

→ The portfolio is structurally stable but highly leveraged. Borrowers are attempting to survive existing debt, meaning new capital is not generating new wealth, increasing systemic fragility.

02

The DTI area chart reveals a steep drop in portfolio safety once a borrower's ratio exceeds 0.26, forming a clear 'Danger Zone' beyond 0.30.

→ DTI is the strongest predictor of contemporary financial strain. Manual underwriting is required for the long-tail demographic exceeding the 0.30 threshold.

03

High-Risk categorization has a near 1:1 correlation (0.991) with a history of bankruptcy, dominating all other negative credit markers.

→ Prior bankruptcies are not anomalous events; they are the primary driver of current portfolio risk and require separate, highly restrictive policy caps.

04

Long-term loans represent the majority of the $526M capital exposure, but the risk funnel shows this capital is safely concentrated in the Low-Risk tier.

→ The core business engine is healthy. The bank can safely continue offering large, long-term facilities to low-risk profiles without adjusting core interest rates.

05

The 'Small Business' loan purpose contains a disproportionately high density of High-Risk applicants relative to its overall volume.

→ Small business lending is the most volatile asset class in the portfolio. Standard consumer risk models are failing here; stricter commercial collateral must be required.

12 & 13. Recommendations & Impact Estimation

Recommendation	Feasibility	Expected Business Impact
Establish 0.30 DTI Hard Ceiling: Automate rejections or require CRO-level sign-off for any applicant with a calculated DTI > 0.30.	High. Easily integrated into the Python ETL and underwriting rules engine.	Risk Reduction: Eliminates the long-tail DTI danger zone, estimated to reduce systemic default probability by 4–6%.
Restrict Bankruptcy Profiles: Cap maximum loan limits and force short-term-only lending for any applicant with a historical bankruptcy flag.	High. Uses existing data points to trigger logic.	Capital Preservation: Directly limits exposure on the demographic with a 0.991 correlation to High Risk, saving millions in potential NPAs.
Direct Disbursement for Consolidation: Require funds for 'Debt Consolidation' to be disbursed directly to creditors, not as cash to the borrower.	Medium. Requires operational adjustments in fund transfers.	Efficiency: Ensures capital is actually used to lower DTI, rather than adding new liquidity to an over-leveraged borrower.

14. Limitations

- Proxy Dependency: The absence of an explicit loan_status field means 'Risk Category' is an engineered heuristic, not a proven empirical outcome.
- Imputation Masking: Applying median values to missing Income and Credit Scores may pull extreme high-risk outliers toward the mean, artificially softening the data.

15. Future Scope

- Predictive Modeling: Upon acquiring historical default outcome data, transition the heuristic Python logic into an XGBoost or Random Forest classification algorithm.
- Macro-Economic Integration: Feed regional housing price indices and inflation rates into the pipeline to contextualize the massive volume of debt consolidation requests.

16. Conclusion

This project successfully transformed raw, unstructured financial records into a clear, actionable risk monitoring system. By deploying an automated Python ETL pipeline to correct anomalies and engineer vital DTI and Risk metrics, the resulting Tableau visualizations decoded a complex $526M portfolio. The analysis proved that while the majority of capital is safely deployed, distinct vulnerabilities exist in the >0.30 DTI band and amongst borrowers with historic bankruptcies. By adopting the recommended hard-ceilings on DTI and bankruptcy caps, the institution can proactively throttle default rates and safeguard long-term capital.

17. Appendix

Data Dictionary

- DTI (Engineered): Debt-to-Income ratio = Monthly Debt / (Annual Income / 12)
- Risk Score (Engineered): Composite count = Bankruptcies + Tax Liens + Credit Problems
- Risk Category (Engineered): Low (0), Medium (1), High (>1)

Python Logic Excerpt

```
# Missing value imputation
df['Credit Score'] = df['Credit Score'].fillna(df['Credit Score'].median())
df['Annual Income'] = df['Annual Income'].fillna(df['Annual Income'].median())
# Handle sentinel values safely
valid_loans = df['Current Loan Amount'].replace(99999999, pd.NA)
loan_upper_limit = valid_loans.quantile(0.99)
df['Current Loan Amount'] = valid_loans.fillna(loan_upper_limit)
# Risk Score
df['Risk Score'] = (
  (df['Number of Credit Problems'] > 0).astype(int) +
  (df['Bankruptcies'] > 0).astype(int) +
  (df['Tax Liens'] > 0).astype(int)
)
```

GitHub Repository

https://github.com/Aman-kumar-git12/D_G14_CreditLoanRiskAnalysis

18. Contribution Matrix

This section must clearly document each team member's contribution across all project phases.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Member 1	✔	✔				✔	
Member 2		✔	✔				✔
Member 3				✔		✔	
Member 4					✔		✔
Member 5	✔				✔		✔
Member 6			✔			✔	

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

Team Lead Signature: _____ Date: _____